

I am preparing a surgical cohort study evaluating the diagnostic performance of CT imaging for detecting extrathyroidal extension (ETE) in thyroid cancer, with surgical pathology as the reference standard. Please identify and summarize PRIOR STUDIES that meet ALL of the following criteria: INCLUSION CRITERIA: 1. Human studies evaluating **CT imaging** (with or without contrast) for detection of **extrathyroidal extension (ETE)** in thyroid cancer 2. **Surgical cohorts** (patients underwent thyroidectomy with pathology confirmation) 3. Report at least ONE of the following diagnostic metrics: - Sensitivity - Specificity - Positive predictive value (PPV) - Negative predictive value (NPV) - Accuracy 4. ETE defined pathologically (gross ETE, microscopic ETE, strap muscle invasion, or beyond strap muscles) EXCLUSION CRITERIA: - Ultrasound-only studies - MRI-only studies - Studies without surgical pathology reference standard - Case reports or series with $n < 50$ - Pediatric-only cohorts FOR EACH STUDY, EXTRACT: - Full citation (authors, journal, year) - Study design (retrospective/prospective) - Sample size (patients and/or nodules) - How CT ETE was defined (radiologist free-text vs structured criteria) - Pathologic ETE definition (gross vs microscopic vs any ETE) - Sensitivity, specificity, PPV, NPV (with confidence intervals if available) - Whether **surgical selection bias** or cohort enrichment was acknowledged SYNTHESIS QUESTIONS: 1. What is the typical **sensitivity range** of CT for detecting ETE? 2. How does CT performance differ for **gross ETE vs microscopic/minimal ETE**? 3. How does use of **strict vs liberal CT criteria** affect sensitivity and specificity? 4. Do prior studies acknowledge **surgical selection bias**, and if so, how? 5. How do these findings contextualize a study showing **low sensitivity but very high specificity** when using conservative/structured CT criteria? Please present results in: - A brief narrative synthesis (2–3 paragraphs) - A comparison table of 3–5 exemplar studies - Bullet-point takeaways suitable for a Discussion section in a clinical manuscript

Prior studies demonstrate that CT imaging for ETE detection in thyroid cancer consistently achieves high specificity (85–98%) but variable sensitivity (14–99%, typically 40–78% with structured criteria), with substantially better performance for gross ETE than microscopic extension, supporting CT's role as a rule-in rather than rule-out tool for preoperative surgical planning.

Abstract

This systematic review synthesized 12 studies evaluating CT diagnostic performance for extrathyroidal extension (ETE) in thyroid cancer surgical cohorts. CT sensitivity for ETE detection varied substantially (14%–99%) depending on the anatomical structure assessed, criteria stringency, and ETE subtype. Specificity was consistently high (85%–98%) when structured CT criteria were employed. CT demonstrated superior performance for gross ETE (AUC 0.800, sensitivity 80%) compared to microscopic ETE (AUC 0.722, sensitivity 50%), reflecting the inherent limitations of cross-sectional imaging for detecting subtle perithyroid soft tissue involvement. Performance also varied by anatomical structure, with vascular invasion showing higher sensitivity (75%–83%) than esophageal invasion (29%–77%). Studies using conservative structured criteria consistently demonstrated the specificity-sensitivity trade-off: very high specificity (>95%) with modest sensitivity (40%–50%), supporting CT's utility for ruling-in rather than ruling-out ETE.

A critical methodological limitation across this literature is inadequate attention to surgical selection bias; fewer than one-third of studies explicitly acknowledged this concern, and no study discussed verification bias. The meta-analysis found over half of included primary studies at high risk of bias. Findings of low sensitivity with very high specificity using conservative CT criteria are well-supported by this evidence base and represent an appropriate diagnostic profile for surgical planning, where confident identification of ETE justifies extended resection while negative findings should not preclude intraoperative vigilance. Emerging radiomics-based approaches (AUC 0.78–0.84) may improve sensitivity without sacrificing specificity, though prospective multicenter validation is needed.

Paper search

We performed a semantic search using the query "I am preparing a surgical cohort study evaluating the diagnostic performance of CT imaging for detecting extrathyroidal extension (ETE) in thyroid cancer, with surgical pathology as the reference standard.

Please identify and summarize PRIOR STUDIES that meet ALL of the following criteria:

INCLUSION CRITERIA:

1. Human studies evaluating **CT imaging** (with or without contrast) for detection of **extrathyroidal extension (ETE)** in thyroid cancer
2. **Surgical cohorts** (patients underwent thyroidectomy with pathology confirmation)
3. Report at least ONE of the following diagnostic metrics:
 - Sensitivity
 - Specificity
 - Positive predictive value (PPV)
 - Negative predictive value (NPV)
 - Accuracy
4. ETE defined pathologically (gross ETE, microscopic ETE, strap muscle invasion, or beyond strap muscles)

EXCLUSION CRITERIA:

- Ultrasound-only studies
- MRI-only studies
- Studies without surgical pathology reference standard
- Case reports or series with $n < 50$
- Pediatric-only cohorts

FOR EACH STUDY, EXTRACT:

- Full citation (authors, journal, year)
- Study design (retrospective/prospective)
- Sample size (patients and/or nodules)
- How CT ETE was defined (radiologist free-text vs structured criteria)
- Pathologic ETE definition (gross vs microscopic vs any ETE)
- Sensitivity, specificity, PPV, NPV (with confidence intervals if available)
- Whether **surgical selection bias** or cohort enrichment was acknowledged

SYNTHESIS QUESTIONS:

1. What is the typical **sensitivity range** of CT for detecting ETE?
2. How does CT performance differ for **gross ETE vs microscopic/minimal ETE**?
3. How does use of **strict vs liberal CT criteria** affect sensitivity and specificity?
4. Do prior studies acknowledge **surgical selection bias**, and if so, how?
5. How do these findings contextualize a study showing **low sensitivity but very high specificity** when using conservative/structured CT criteria?

Please present results in:

- A brief narrative synthesis (2–3 paragraphs)
- A comparison table of 3–5 exemplar studies
- Bullet-point takeaways suitable for a Discussion section in a clinical manuscript” across over 138 million academic papers from the Elicit search engine, which includes all of Semantic Scholar and OpenAlex.

We retrieved the 500 papers most relevant to the query.

Screening

We screened in sources based on their abstracts that met these criteria:

- **Human Thyroid Cancer Studies:** Does this study involve human patients with thyroid cancer (excluding animal or in vitro studies)?
- **CT Imaging for ETE Detection:** Does this study evaluate CT imaging for detection of extrathyroidal extension (ETE) in thyroid cancer?
- **Surgical Pathology Reference Standard:** Does this study include surgical cohorts with thyroidectomy and histopathological confirmation as the reference standard for ETE assessment?
- **Quantitative Diagnostic Performance Data:** Does this study report quantitative diagnostic performance metrics (sensitivity, specificity, positive predictive value, negative predictive value, or accuracy) with sufficient data to extract or calculate these metrics?
- **Adequate Sample Size:** Does this study include 50 or more patients or nodules in the analysis?
- **Pathologically Defined ETE:** Does this study use pathologically defined extrathyroidal extension as the outcome measure?
- **Full-Text Publication:** Is this a full-text publication (not just a conference abstract without accompanying full publication)?
- **Adult Population:** Does this study include adult patients (not exclusively pediatric populations)?

We considered all screening questions together and made a holistic judgement about whether to screen in each paper.

Data extraction

We asked a large language model to extract each data column below from each paper. We gave the model the extraction instructions shown below for each column.

- **Study Characteristics:**

Extract basic study information including:

- Full citation (first author, journal, year)
- Study design (retrospective vs prospective cohort)
- Study period/dates
- Sample size (number of patients AND number of nodules/lesions if different)
- Setting (single center vs multicenter, country)

- **CT Methodology:**

Extract details about CT imaging approach including:

- CT scanner type and technique (contrast enhanced vs non-contrast)
- How ETE was assessed on CT (radiologist free-text interpretation vs structured criteria vs scoring system)
- CT criteria used to define ETE presence (e.g., loss of fat plane, tissue invasion, specific anatomical involvement)
- Whether strict/conservative vs liberal/broad CT criteria were used
- Number and experience level of radiologists interpreting CT
- Blinding status of CT readers to pathology results

- **Pathology Reference Standard:**

Extract pathological ETE definition and assessment including:

- How ETE was defined pathologically (gross ETE only vs microscopic ETE vs any ETE vs specific structures invaded)
- Whether minimal/microscopic ETE was distinguished from gross/extensive ETE
- Specific anatomical structures considered for ETE (strap muscles, trachea, esophagus, vessels, etc.)
- Pathologist experience/specialization
- Whether pathologists were blinded to CT findings

- **Diagnostic Performance:**

Extract all reported diagnostic metrics for CT detection of ETE including:

- Sensitivity (%) with 95% confidence intervals if provided
- Specificity (%) with 95% confidence intervals if provided
- Positive predictive value (PPV) (%) with confidence intervals
- Negative predictive value (NPV) (%) with confidence intervals
- Accuracy/diagnostic accuracy (%) with confidence intervals
- Area under ROC curve (AUC) if provided
- Any other performance metrics reported

- **ETE Subgroup Analysis:**

Extract any subgroup analyses of CT performance by ETE type or severity including:

- Diagnostic metrics for gross/extensive ETE vs microscopic/minimal ETE separately

- Performance by specific anatomical structure invaded (strap muscle vs trachea vs major vessels)
- Performance by tumor size categories
- Performance by histological subtype if reported
- Any discussion of why performance differs by ETE subtype

- **Selection Bias Assessment:**

Extract information about surgical selection bias and study limitations including:

- Whether authors acknowledge surgical selection bias or cohort enrichment
- Discussion of verification bias (whether all patients with negative CT underwent surgery)
- Proportion of patients with ETE in the surgical cohort vs expected population prevalence
- Any discussion of how patient selection for surgery might affect diagnostic performance
- Other methodological limitations acknowledged by authors

- **Key Findings:**

Extract the main conclusions and clinical implications including:

- Authors' primary conclusions about CT performance for ETE detection
- Clinical recommendations for CT use in ETE assessment
- Comparison to ultrasound performance if included
- Any discussion of optimal CT criteria or thresholds
- Authors' assessment of CT's clinical utility for surgical planning

Characteristics of Included Studies

Twelve studies met the inclusion criteria for this systematic review, encompassing both primary diagnostic accuracy studies and one meta-analysis. The majority employed retrospective cohort designs, with sample sizes ranging from 83 to 624 patients in primary studies, while the meta-analysis incorporated 31,942 observations across 47 studies.

Study	Full Text Retrieved?	Study Type	Design	Sample Size	Setting	CT Technique
Y. Seo et al., 2010	No	Primary study	Retrospective	84 patients, 86 tumors	Not specified	Not specified
Abhishek Mahajan et al., 2024	No	Primary study	Retrospective	83 patients	Single center	Structured criteria (Mahajan grading)
Muhammad Asad Iqbal et al., 2025	No	Primary study	Retrospective	102 patients, 136 nodules	Single center, China	Dual-source dual-energy CT
Trương Thụy My et al., 2023	Yes	Primary study	Retrospective/prospective cross-sectional	156 patients, 116 tumors	Single center, Vietnam	Contrast-enhanced CT
Doh Young Lee et al., 2014	No	Primary study	Retrospective	377 patients	Not specified	Not specified

Study	Full Text Retrieved?	Study Type	Design	Sample Size	Setting	CT Technique
J. Choi et al., 2009	No	Primary study	Prospective cohort	299 patients	Not specified	Not specified
Mostafa Alabousi et al., 2021	No	Systematic review/meta-analysis	Meta-analysis	31,942 observations	Multiple centers	Various
Jee-Hye Choi et al., 2022	No	Primary study	Retrospective	424 patients	Not specified	Not specified
Bin Chen et al., 2019	Yes	Primary study	Retrospective	624 patients	Single center, China	Contrast-enhanced CT with structured criteria
Pengyi Yu et al., 2022	Yes	Primary study	Retrospective	199 patients (multicenter)	Multicenter, China	Contrast-enhanced CT with structured criteria
Y. Choi et al., 2010	No	Primary study	Not specified	589 patients	Not specified	Not specified
X. Xu et al., 2022	Yes	Primary study	Retrospective	452 patients	Single center, China	Third-generation dual-energy CT

The included studies predominantly originated from Asian centers, with several conducted in China and Vietnam . Study periods ranged from 2010 to 2023, reflecting evolution in CT technology and diagnostic approaches over time. Four studies had full text available for detailed extraction , while the remainder were limited to abstract-level data.

CT Methodology and ETE Assessment Criteria

CT methodological approaches varied substantially across studies. Several studies employed advanced dual-energy CT techniques , while others used conventional contrast-enhanced protocols . The assessment of ETE on CT images ranged from unstructured radiologist interpretation to structured grading systems.

Study	CT Assessment Method	Specific ETE Criteria	Radiologists	Blinding Status
Y. Seo et al., 2010	Two radiologists, independent evaluation	Not specified	2 radiologists	Not specified

Study	CT Assessment Method	Specific ETE Criteria	Radiologists	Blinding Status
Abhishek Mahajan et al., 2024	Structured Mahajan grading system	Graded into categories for ETE and tracheal invasion	2 radiologists	Not specified
Muhammad Asad Iqbal et al., 2025	Quantitative parameters (iodine concentration, spectral curve slope)	Threshold-based (e.g., iodine concentration 2.88 mg/ml)	Not specified	Not specified
Trương Thụy My et al., 2023	Experienced radiologists using window changes and thin-slice reconstruction	Irregular margins, unclear boundaries, calcification, contrast enhancement	Experienced radiologists	Not specified
Bin Chen et al., 2019	Structured criteria based on prior studies	Contact with thyroid capsule >25%, contact with tracheal/esophageal/vascular structures, loss of normal structures	2 radiologists with >10 years experience	Not specified
Pengyi Yu et al., 2022	Structured criteria	Tumor contact $\geq 180^\circ$ with trachea/esophagus/muscle/capsule, loss of normal structure, focal capsule disruption, >25% perimeter abutting capsule	2 radiologists (15 and 5 years experience)	Blinded to histopathology
X. Xu et al., 2022	Structured criteria by experienced radiologists	Criteria summarized in supplemental data	3 radiologists (6, 5, and 8 years experience)	Blinded to clinical data and pathology

Studies employing structured CT criteria consistently demonstrated higher specificity but variable sensitivity, reflecting the trade-off inherent in conservative diagnostic thresholds .

Pathologic Reference Standard Definitions

The pathologic definition of ETE varied across studies, with some distinguishing between gross and microscopic extension while others combined both categories.

Study	ETE Definition	Gross vs Microscopic Distinction	Anatomical Structures Considered	Pathologist Blinding
Y. Seo et al., 2010	T3 and T4 extracapsular extension	Not distinguished	Trachea, esophagus, carotid artery, jugular vein, RLN	Not specified
Trương Thụy My et al., 2023	Not explicitly distinguished	Not distinguished	Strap muscles, trachea, esophagus, blood vessels	Not specified
Jee-Hye Choi et al., 2022	Gross ETE	Focused on gross ETE	Not specified	Not specified
Bin Chen et al., 2019	Both minimal and gross ETE per AJCC TNM	Distinguished but combined for analysis	Strap muscles, trachea, larynx, vasculature, esophagus, RLN	Not specified
Pengyi Yu et al., 2022	AJCC 8th edition (gross and microscopic)	Distinguished per AJCC definition	Strap muscles, trachea, vasculature, larynx, esophagus, RLN	Not specified
X. Xu et al., 2022	Histopathologic confirmation, including microscopic ETE	Microscopic ETE considered	Not detailed	Not specified

The heterogeneity in pathologic ETE definitions represents a significant source of variability in reported diagnostic performance metrics, as microscopic extension is inherently more challenging to detect on cross-sectional imaging than gross invasion of adjacent structures.

Diagnostic Performance of CT for ETE Detection

Overall Performance Metrics

The diagnostic performance of CT for ETE detection demonstrated consistent patterns across studies: high specificity with variable sensitivity depending on the anatomical structure assessed and the stringency of CT criteria employed.

Study	Sensitivity	Specificity	PPV	NPV	Accuracy	AUC
Y. Seo et al., 2010 (trachea)	59.1%	91.4%	Not reported	Not reported	83.2%	Not reported
Y. Seo et al., 2010 (esophagus)	28.6%	96.2%	Not reported	Not reported	90.7%	Not reported

Study	Sensitivity	Specificity	PPV	NPV	Accuracy	AUC
Y. Seo et al., 2010 (carotid artery)	75.0%	99.4%	Not reported	Not reported	98.8%	Not reported
Y. Seo et al., 2010 (jugular vein)	33.3%	98.8%	Not reported	Not reported	97.1%	Not reported
Y. Seo et al., 2010 (RLN)	78.2%	89.8%	Not reported	Not reported	85.5%	Not reported
Abhishek Mahajan et al., 2024 (overall ETE)	14.3%–77.8%	95%–98%	Not reported	Not reported	82%–87%	Not reported
Abhishek Mahajan et al., 2024 (tracheal invasion)	78%	75%	Not reported	Not reported	91.5%	Not reported
Trương Thụy My et al., 2023 (capsular invasion)	99%	84.6%	98.1%	91.7%	97.4%	Not reported
Trương Thụy My et al., 2023 (strap muscles)	98.7%	97.4%	98.7%	97.4%	98.2%	Not reported
Trương Thụy My et al., 2023 (trachea)	77.3%	98.9%	94.4%	94.9%	94.8%	Not reported
Trương Thụy My et al., 2023 (esophagus)	76.5%	98.9%	92.9%	96.1%	95.7%	Not reported
Trương Thụy My et al., 2023 (blood vessels)	83.3%	100%	100%	99%	99.1%	Not reported
Jee-Hye Choi et al., 2022 (gross ETE)	75.9%	Not reported	Not reported	Not reported	Not reported	Not reported
Bin Chen et al., 2019	47.7%	87.9%	Not reported	Not reported	Not reported	0.772–0.812
Pengyi Yu et al., 2022	69.2% (95% CI 38.9%–89.6%)	90.9%	Not reported	Not reported	84.8%	0.750–0.797

Study	Sensitivity	Specificity	PPV	NPV	Accuracy	AUC
X. Xu et al., 2022	40.2%	97.4%	Not reported	Not reported	Not reported	0.78–0.84

The data reveal substantial heterogeneity in sensitivity values, ranging from 14.3% to 99% depending on the study methodology and target structure . Specificity consistently exceeded 85% across most studies, with several reporting values above 95% .

Performance by Anatomical Structure

CT diagnostic performance varied considerably by the anatomical structure being assessed. Vascular invasion demonstrated higher sensitivity (75%–83.3%) compared to esophageal invasion (28.6%–76.5%) . Tracheal invasion detection showed moderate sensitivity ranging from 59.1% to 78% . The recurrent laryngeal nerve invasion had relatively high sensitivity at 78.2% , though this finding requires confirmation in additional studies.

Gross versus Microscopic ETE Performance

Studies that distinguished between gross and microscopic ETE demonstrated predictably better performance for gross extension. One study reported successful identification of 85.4% of patients with minimal ETE and 80.6% with gross ETE, though detailed sensitivity and specificity metrics were not provided separately . Another study examining gross versus microscopic ETE subgroups found AUC values of 0.800 for gross ETE versus 0.722 for microscopic ETE, with corresponding sensitivities of 80.0% and 50.0% respectively .

Radiomics-Enhanced Performance

Three studies evaluated radiomics-based approaches to improve CT diagnostic performance . The radiomic nomogram developed by Bin Chen et al. achieved an AUC of 0.812 in the validation cohort, significantly outperforming conventional radiologist interpretation . Similarly, Pengyi Yu et al. demonstrated AUC values of 0.750–0.797 with their radiomics nomogram , while X. Xu et al. achieved AUC of 0.78–0.84 using iodine map-based radiomics . These studies suggest that quantitative image analysis may enhance conventional CT interpretation for ETE detection.

Comparison with Ultrasound

Several studies directly compared CT and ultrasound performance. One study found ultrasound more accurate than CT for predicting ETE overall , while another demonstrated CT has higher sensitivity than ultrasound for evaluating invasive thyroid cancer . The meta-analysis by Alabousi et al. reported ultrasound sensitivity of 91% and specificity of 47% for extrathyroidal extension , though corresponding CT-specific metrics for ETE were not provided. The clinical recommendation emerging from these comparisons suggests ultrasound as a first-line modality for minimal ETE, with CT reserved for evaluating maximal ETE and central lymph node metastasis .

Selection Bias Assessment

Assessment of surgical selection bias was inconsistently addressed across included studies.

Study	Selection Bias Acknowledged	Verification Bias Discussed	Other Limitations
Y. Seo et al., 2010	Not mentioned	Not mentioned	Low sensitivity acknowledged Not discussed
Abhishek Mahajan et al., 2024	Not mentioned	Not mentioned	Not discussed
Muhammad Asad Iqbal et al., 2025	Not mentioned	Not mentioned	Not discussed
Trương Thụy My et al., 2023	Not explicitly acknowledged	Not discussed	Patients without complete surgical reports excluded
Doh Young Lee et al., 2014	Not mentioned	Not mentioned	Not discussed
J. Choi et al., 2009	Not mentioned	Not mentioned	Consecutive registration noted
Mostafa Alabousi et al., 2021	QUADAS-2 risk of bias assessment performed	Not explicitly discussed	26/47 studies at high risk of bias
Jee-Hye Choi et al., 2022	Need for careful patient selection acknowledged	Not discussed	Selection bias implicitly recognized
Bin Chen et al., 2019	Acknowledged potential selection bias from exclusion criteria	Not discussed	Single-institution limitation, small/indistinct lesion exclusion
Pengyi Yu et al., 2022	Retrospective design selection bias acknowledged	Not discussed	Small PTCs potentially underrepresented
Y. Choi et al., 2010	Not mentioned	Not mentioned	Not discussed
X. Xu et al., 2022	Single-center retrospective design acknowledged	Not discussed	Radiation exposure, contrast limitations noted

Only four studies explicitly acknowledged potential selection bias . The meta-analysis applied formal bias assessment, finding 26 of 47 included studies at high risk of bias . Notably, no study explicitly discussed verification bias (whether all patients with negative CT findings proceeded to surgery), which represents a critical methodological consideration in surgical cohort studies evaluating diagnostic accuracy.

Synthesis

Typical Sensitivity Range for CT Detection of ETE

The sensitivity of CT for ETE detection demonstrates marked heterogeneity, ranging from approximately 14% to 99% across included studies . This wide range reflects several sources of variability including the anatomical structure assessed, the stringency of CT criteria employed, and whether gross or microscopic ETE was the target. For conventional radiologist interpretation using structured criteria, sensitivity typically falls between 40% and 78% . Studies reporting very high sensitivity (>90%) generally assessed capsular invasion or employed less stringent diagnostic criteria .

Differential Performance for Gross versus Microscopic ETE

The available evidence indicates substantially better CT performance for gross ETE compared to microscopic extension. One study demonstrated AUC of 0.800 for gross ETE versus 0.722 for microscopic ETE, with sensitivity of 80% versus 50% respectively . This finding is mechanistically predictable, as microscopic tumor extension into perithyroid soft tissues lacks the conspicuous imaging features (tissue distortion, loss of fat planes, organ displacement) that characterize gross invasion . The distinction is clinically relevant because microscopic ETE, while pathologically significant, may not alter surgical approach in the same manner as gross invasion into trachea, esophagus, or major vessels .

Impact of Strict versus Liberal CT Criteria

Studies employing conservative or structured CT criteria consistently demonstrate a specificity-sensitivity trade-off. Conservative criteria—such as requiring >25% tumor-capsule contact, $\geq 180^\circ$ circumferential involvement of adjacent structures, or definite loss of tissue planes—yield very high specificity (typically >95%) at the expense of sensitivity (40–50%) . In contrast, studies using more liberal interpretive criteria report higher sensitivity but reduced specificity . This pattern supports the rule-in versus rule-out paradigm: conservative CT criteria are appropriate when high confidence in positive findings is required for surgical planning, accepting that some cases of ETE will be missed and discovered at pathology.

Acknowledgment of Surgical Selection Bias

The majority of included studies did not explicitly address surgical selection bias . Among those acknowledging limitations, the discussion typically focused on retrospective design and single-center limitations rather than the specific bias introduced by studying only patients who underwent surgery . This represents a significant gap, as surgical cohorts likely enrich for patients with clinically suspicious disease, potentially inflating both the pretest probability of ETE and the apparent specificity of imaging. The meta-analysis finding that over half of included studies were at high risk of bias underscores this concern .

Contextualizing Low Sensitivity with High Specificity Findings

A study demonstrating low sensitivity but very high specificity when using conservative CT criteria aligns with the preponderance of evidence from this systematic review. Multiple studies confirm this pattern: X. Xu et al. reported 40.2% sensitivity with 97.4% specificity ; Bin Chen et al. found 47.7% sensitivity with 87.9% specificity ; and Mahajan et al. documented sensitivity as low as 14.3% with specificity up to 98% .

This diagnostic profile has clear clinical implications. When CT demonstrates definite ETE using structured criteria, surgical planning can confidently proceed with extended resection approaches, as the high specificity indicates minimal false-positive risk. Conversely, the low sensitivity mandates that absence of CT-apparent ETE should not provide false reassurance; surgeons must maintain vigilance for intraoperative findings of extension. The radiomics-enhanced approaches showing improved AUC (0.78–0.84) suggest potential for improving sensitivity without sacrificing specificity, though these require validation in prospective multicenter studies.

Key Takeaways for Discussion

- CT sensitivity for ETE detection ranges widely (14%–99%) depending on the anatomical structure assessed, criteria stringency, and whether gross or microscopic ETE is the target

- Specificity is consistently high (typically >85%–98%) when structured CT criteria are employed, supporting CT's role in confidently identifying positive cases for surgical planning
- Gross ETE demonstrates significantly better CT detection than microscopic ETE, with approximately 30% difference in sensitivity between these subgroups
- Performance varies by anatomical structure: vascular invasion (75%–83% sensitivity), tracheal invasion (59%–78%), and esophageal invasion (29%–77%) show different detection rates
- Radiomics-based approaches improve diagnostic performance (AUC 0.78–0.84) compared to conventional radiologist interpretation and may represent a future direction for CT-based ETE assessment
- Surgical selection bias was acknowledged in fewer than one-third of studies, representing a systematic methodological limitation in this literature
- The combination of US and CT may provide optimal preoperative staging, with US preferred for minimal ETE and CT for maximal ETE and central lymph node assessment
- Low sensitivity with high specificity findings using conservative CT criteria are consistent with the existing literature and represent an appropriate trade-off for surgical planning purposes where ruling-in disease carries greater clinical significance than ruling-out

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